

HaptoRelax: Haptic Feedback for Enhancing Human Touch in VR Relaxation Using SMA

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Figure 1: We created HaptoRelax, an SMA-actuated wearable haptic device for human touch in VR relaxation. (a) The device structure and its mechanism to generate repetitive grabbing. (b) Users wear HaptoRelax on their wrist and put on VR HMD to have a relaxing multisensory experience. (c) Users can have a relaxing time with their friends or family in VR scenes not only visually, but also physically feeling the sensation of gentle touch from each other.

ABSTRACT

Stress management is a growing concern in modern society and virtual reality (VR) is emerging as a promising tool to alleviate stress. While most VR relaxation applications rely on audio-visual cues, incorporating haptic sensations, particularly gentle human touch, can enhance the immersive experience and effectiveness of stress reduction. This research paper introduces HaptoRelax, a wrist-worn device utilizing shape memory alloys (SMA) to provide soft and smooth tactile feedback on the skin. We used three-shape memory alloy(SMA) springs to generate a gentle repetitive grabbing. Our conducted user study reveals that our haptic device maps to the sensation of wrist grabbing, and can be helpfully applied within VR relaxation scenes. .

Index Terms: Virtual Reality (VR)—Shape Memory Alloy SMA—Relaxation techniques—VR Relaxation; Human Touch—Touch Simulation—Haptic Device—Haptic Feedback

1 INTRODUCTION

Stress is a ubiquitous issue in modern society, and individuals are increasingly seeking effective methods to manage it. Virtual reality (VR) is emerging as a promising new approach to stress management, as it allows easier access to activities such as meditation, reflection, and mindfulness. However, most VR relaxation applications rely on audio-visual cues, such as soothing scenes and calming music, to promote relaxation [7, 23]. It is already known that haptic sensations, such as human touch, can enhance overall well-being by reducing stress [2, 5]. Incorporating human touch into the VR environment is crucial to provide a more immersive and realistic experience for users. Recent works show that VR has the potential to positively impact emotional well-being and relaxation [8, 16].

Furthermore, the VR environment has the potential to be a place for reunion among families, friends, and couples by enabling remote communication in an immersive manner. However, despite the potential, there is currently no approach that effectively renders gentle human touch haptic feedback to elicit positive emotions [12]. Various actuation methods for generating human touch haptic feedback have been explored in the field of human-computer interaction (HCI). Pneumatic devices have been used, but they often require bulky hardware setups [13]. Cable-driven methods, on the other hand, tend to actuate in an unnatural way compared to real gentle human touch [24].

In our approach, we focus on utilizing shape memory alloys (SMA) to render soft and smooth tactile feedback on the skin [15, 18, 21]. SMAs have shown potential in simulating human touch sensations. Specifically, we have developed a wrist-worn device that utilizes the structure of SMAs, inspired by the work of [9, 10], to create repetitive grabbing motions on the wrist. By choosing the wrist as the targeted body part, we aim to provide a natural and comfortable haptic experience for users.

By integrating haptic feedback and specifically focusing on gentle human touch sensations, our device aims to enhance the effectiveness of VR for stress management and relaxation. Our research takes the contributions in 1) introducing which parameters affect the relaxing experience in VR with SMA-actuated grabbing sensations, and 2) the structure of the device rendering gentle and repetitive grabbing.

2 RELATED WORK

2.1 Importance of Human Touch

Numerous studies emphasize the profound significance of human touch in fostering positive emotions and alleviating stress. Extensive research has consistently demonstrated that touch possesses the remarkable ability to enhance overall well-being by effectively reducing stress levels [2, 5]. In therapeutic practices, touch assumes a fundamental role in human communication, enabling individuals to regulate their emotions, alleviate anxiety, establish connections with others, uplift mood, and promote deep relaxation [4, 6, 14]. Furthermore, a notable discovery by [12] reveals that different forms of touch, including patting, stroking, and rubbing, have the power

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to evoke diverse emotions and instill empathy. Consequently, leveraging virtual reality (VR) to simulate the sensation of human touch emerges as a promising avenue for delivering a more authentic and tactile relaxation experience.

2.2 Human Touch Haptic Devices

Various methods are employed to create tactile sensations on the skin. Rotational actuators with vibrotactors connected via cords around the wrist are used to generate squeezing and vibrotactile sensations [20]. While capable of providing squeezing feedback, these approaches primarily focus on the mechanism rather than creating soft and relaxing sensations.

Vibrotactile motors have been utilized in interpersonal communication for affective haptic feedback [13, 19, 24]. However, these methods suffer from drawbacks such as bulkiness, noticeable noise, and limited bandwidth, which hinder the achievement of gentle human touch sensations.

To address these limitations, researchers have explored pneumatic actuators. A fabric-based sleeve controlled by pneumatic pressure has been developed to render a wide range of sensations for augmented reality (AR) and virtual reality (VR) applications [25]. Another approach involves a pneumatic armband capable of generating tapping, holding, and tracing sensations on the wrist [11]. However, these methods require bulky air pressures, limiting their practicality in everyday scenarios.

2.3 SMA-based Interface

SMA is an alloy that is malleable in a room temperature but returns to its pre-trained shape with a strong force when heated until its transition temperature. [19] investigates how rhythmic haptic action of warmth and slight pressure along the arm by an SMA-based sleeve can promote calmness and reduce anxiety. However, the research handles the haptic feedback as a simple pattern rather than a repetitive human touch on the wrist.

As SMA actuates in a quiet, gentle and compact manner, many HCI research utilize SMAs to replicate smooth human touch haptic sensations. [10] presents thin and flexible sticker-type tactile interfaces on the skin, replicating the sensations of pinching, directional stretching, pressing, pulling, dragging, and expanding. [18] renders sensations of grabbing, tapping, stroking and encircling on the forearm by attaching zigzag stitched SMA patches in a matrix on the forearm. [9] investigates squeeze sensations by covering the wrist with SMA springs. [15, 21] uses SMA wires connected with rigid structures that can make different sensations on the skin, such as tickling, pinching, squeezing and twisting on the wrist. These previous works present the potential of SMA as a haptic device for rendering human touch. However, none of the research focuses on how this human touch haptic feedback can be expanded to elicit a relaxed state in conjunction with soothing VR scenes.

Furthermore, [19] suggests the pace of the haptic action can influence the participants' perception of calmness, it is challenging to develop an SMA-based interface capable of repetitive motions. This is because manual elongation or deformation of the SMA is needed once it has returned to its pre-trained form after heated. For these reasons, prior works do not handle repetitive feedback or manually deform SMAs every cycle, except [18] which only requires a cooling process. Therefore, we build on the actuation mechanisms of using multiple SMA springs from [9, 10] to make human touch-grabbing motion repeatable.

3 HAPTORELAX

HaptoRelax consists of three SMA springs, powered in turns to generate repetitive grabbing haptic feedback on the wrist. In order to choose the best structure, we go through material selection, an iterative design process, and lastly the hardware setup.

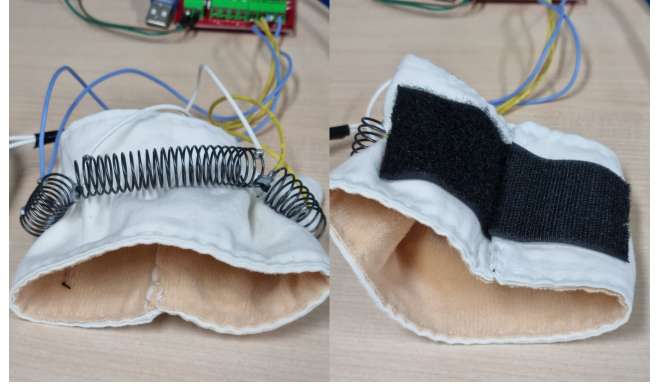


Figure 2: Final Structure of HaptoRelax. A long SMA spring (50.5mm) is connected with shorter springs (20.0mm) on each end. When the long spring is supplied with power, it contracts making the other two springs to elongate. After that, the shorter two springs are supplied with power simultaneously, elongating the long spring. (left) Velcro strap is used so that the wrist band can fit to people with different wrist circumferences. (right)

Table 1: Material Selection

SMA Product (Supplier)	Diameter (mm)	Transition Temperature (°C)	Actuation Time (sec)
Nitinol Wire (NexMetal)	0.5	70	~6
Nitinol Wire (NexMetal)	1	40	~60
Nitinol Wire (KRL)	1.0	80	~60
Nitinol Spring (KRL)	1.0	45	~7

3.1 Material Selection

We conducted a comparison of four types of shape memory alloys (SMAs) in terms of actuation time and perceived temperature during actuation. We used: 1) $\phi 0.5\text{mm}$ NexMetal with a transition temperature of 70°C , 2) $\phi 1\text{mm}$ NexMetal with a transition temperature of 40°C , and 3) $\phi 1\text{mm}$ Kellogg's Research Lab (KRL) wire with a transition temperature of 80°C (Table 1). All wires were cut into a uniform length of 20mm, deformed into a helical shape, and connected to a 5V DC power supply (RDP-303AU, SMART).

The actuation time was the shortest for the $\phi 0.5\text{mm}$ wire, taking approximately 6 seconds to return to the pre-trained state of a straight wire. In contrast, both $\phi 1\text{mm}$ wires required around 1 minute for actuation. The perceived temperature on bare skin during actuation was lowest with the $\phi 0.5\text{mm}$ wire, while the $\phi 1\text{mm}$ wires were unsuitable due to the risk of burns. This was attributed to the significantly longer actuation times of the $\phi 1\text{mm}$ wires, as the duration of current flow through the wire led to increased heat energy.

Additionally, we used a $\phi 1\text{mm}$ KRL SMA spring with a transition temperature of 45°C . The spring had a coil diameter of 14mm and consisted of 24 active coils. It took approximately 7 seconds for the spring to return to its original tight pitch after elongation by an external force. The perceived heat was higher compared to the $\phi 0.5\text{mm}$ wire due to the same reason as the $\phi 1\text{mm}$ wire.

Based on our evaluation, we selected the Nitinol Spring for our device due to its quick actuation and sufficient contraction force, providing a human touch-like sensation.

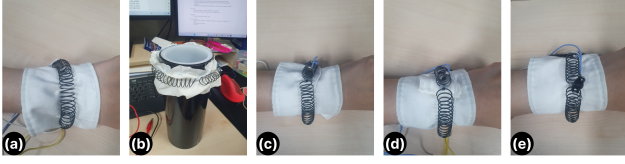


Figure 3: Iterative design process. (a) Single spring (50.5mm), (b) Two springs (each 50.5mm), (c) Two springs (50.5mm & 20.0mm), (d) Two springs connected with fabric (50.5mm & 20.0mm), and (e) Two springs connected with 3-D printed sphere (50.5mm & 20.0mm)

3.2 Design Iteration

According to [9], commercially available SMA wires do not provide enough load for satisfactory squeezing sensations and require a spring for restoration. Our iterative design process aims to find an effective structure of SMA springs that can exert detectable grabbing sensations without relying on external force for restoration.

Initially, we used a single long spring wrapped around the whole wrist (Figure 3a). While the grabbing force and sensation was sufficient, the structure was not capable of restoration. To address this, we experimented with two connected 50.5mm springs (Figure 3b). This allowed for repetitive movements, but the total length of the springs was too long to provide a comfortable grabbing sensation for the human wrist.

To overcome this, we combined a shorter 20.0mm spring with the original longer spring (Figure 3c). The longer spring served as the main actuator, providing adequate grabbing force, while the shorter spring facilitated the elongation of the longer spring. By alternately powering the springs, repetitive motions were made possible. To enhance the sensation of the springs elongating and contracting, we used fabric and 3-D printed spheres for connection (Figure 3d, e). However, we observed that the shape and material of the connecting parts had less impact compared to the number and placement of the springs and the power supply.

In the final design (Figure 2), a single long spring with two shorter springs connected at each end was chosen. This configuration provided a stronger grabbing force on the wrist, enabling repetitive grabbing sensations by alternating the power supply to the long and short springs. To ensure a relaxing haptic feedback experience, the springs were attached to two layers of textiles: cotton and polyester. Cotton was used to prevent the risk of polyester fabric burning when in direct contact with the SMAs. Soft polyester fabric was utilized for skin contact, following the findings in [22], highlighting the potential of soft fabric textures in creating effective contact with the skin.

In using SMA for gentle human touch, actuating and cooling speed, contraction force, temperature, and current should be considered. Moreover, various approaches to embedding SMA into the device can generate different motions. The order of spring actuation also influences the perception. It can be further developed to render even stroking.

3.3 Current Calculation

We calculated the current needed for actuating SMA. The total length of each spring was 1142.96mm, 483.56mm, and 439.6mm, which was obtained by the Eq. 1. (Coil Diameter $D_m = 14\text{mm}$, Number of Active Coils $i = 24, 9, 8$ for each spring)

$$L = \pi D_m i + 2\pi D_m \quad (1)$$

The resistivity of the Nitinol is *resistivity of Nitinol* $\rho = 7.6 \times 10^{-2} \Omega \cdot \text{mm}$, leading to the resistance of each spring 1.106 Ω , 0.468 Ω , and 0.452 Ω from the Eq. 2.

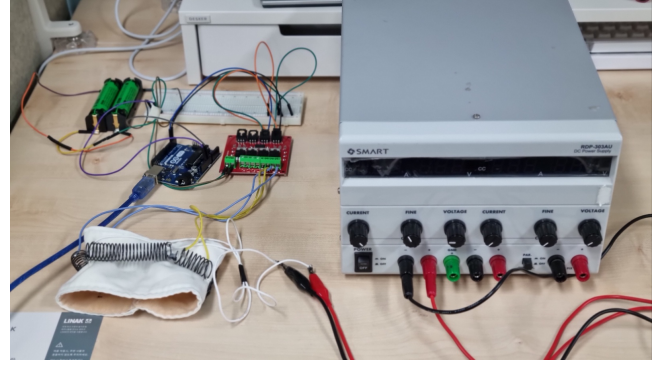


Figure 4: Final HaptoRelax Setting

$$R = \rho L / A \quad (2)$$

The latent heat of transformation and Specific heat of nitinol are each $r = 20\text{J/g}$ and $C = 0.01\text{J} / \cdot \text{gC}$. T_1 is a room temperature of 25C and T_2 is a transition temperature of the spring which is 45C. By Eq. 3, the total energy needed to actuate for each spring is 69.286J, 26.26J, and 23.03J.

$$E = Q_L + Q_S = r \cdot m + (T_2 - T_1) \cdot m \cdot C \quad (3)$$

As $P = E/t = VI = I^2 R$ and setting the actuation time $t = 10\text{sec}$ for the long spring and $t = 7\text{sec}$ for shorter springs, the total power is 6.9W, 3.75W, and 3.29W for each spring. The current needed to actuate is 2.5A, 2.77A, and 2.78A each.

3.4 Hardware Setup

To control the three springs sequentially, we utilized a motor driver (SZH-AT021) connected to two shorter springs. The motor driver was powered by 7.4V batteries and controlled using Arduino Uno. To compensate for power loss when actuating two SMAs simultaneously, we provided the shorter springs with slightly more power than required. The longer springs were not connected to the same motor driver due to the high current they demand, which the DC power supply and driver were unable to handle. Figure 4 illustrates the final setup of HaptoRelax.

4 USER STUDIES

4.1 Preliminary Study on VR Preferences

We conducted a preliminary survey using VR Relaxation to narrow down our focus on a specific body part and evaluate the effectiveness of implementing touch sensations for relaxation purposes. Our quick survey, distributed via Google Forms through platforms like KakaoTalk and WhatsApp, consisted of 20 questions divided into five topic blocks: Demographics (7 questions), technology acceptance (4 questions) (Neyer et al., 2012), importance of touch in stressful situations (3 questions), VR relaxation (4 questions), and preferences on VR relaxation (2 questions).

Our findings indicate that participants experience occasional stress in their daily lives and recognize the significance of touch for relaxation during stressful situations, positively impacting their overall mental well-being. Attitudes towards VR relaxation were ambivalent, suggesting the potential to shape positive opinions. Among the body parts, the back emerged as the most preferred for simulating touch, followed by hands and arms, while no strong preferences were observed for specific touch sensations.

4.2 Absolute Threshold Detection

In order to establish the minimum power required for user's perception of grabbing, we initially carried out a pilot study employing a two-down, one-up staircase method to pinpoint the absolute threshold. For the absolute threshold detection, we varied the voltage supplied to the device using a variable power supply. By a starting power of 3V, we decreased the power in each trial by 0.2V. The number of trials was not set beforehand. We did as many trials as needed to get 3 reversals (three peaks and lows). Each trial lasted 10 seconds followed by a ten-second cooling break between each trial for guaranteeing safety since the SMA springs get hot.

We did 21 trials with six participants in total. The total average absolute threshold was 6.5W.

4.3 User Experience Study

In the following qualitative user experience study the authenticity of HaptoRelax in simulating human touch and its effectiveness in promoting relaxation was assessed, aiming to answer the following questions:

1. How authentic do users perceive HaptoRelax for simulating human touch?

The purpose of this question is to evaluate the perceived realism of the simulated touch sensations. Understanding how users perceive the authenticity of grabbing is crucial for evaluating the effectiveness of the SMA-shape wrist-worn device. If users perceive the sensation as realistic, it enhances the overall user experience. On the contrary, if users do not find it realistic, it may indicate the need for improvements in the device's functionality and design to enhance the realism of grabbing.

2. How effective is HaptoRelax to promote relaxation?

By assessing the device's impact on relaxation, we can determine its potential as a stress reduction tool. If users report a significant improvement in relaxation levels after using the device, it provides evidence of its efficacy and its potential as a valuable tool for managing stress. Conversely, if users do not experience a notable impact on relaxation, it may indicate the need for further refinement or modifications to enhance the device's effectiveness.

This user experience study was divided into two segments, complemented by a screening questionnaire (age, gender, prior experience with VR and relaxation techniques) at the beginning and a post-questionnaire after the testing to evaluate VR-induced symptoms and effects (VRISE). In the first part of the testing, the users' experience of HaptoRelax was conducted without using HMD. In the second part, users were placed in a calm virtual environment where HaptoRelax was in sync with the touch animation.

Both parts were accompanied by eight questions divided into three blocks: First impression (one question), realistic touch sensation (three questions), and relaxation (four questions).

Participants were familiarized with the testing procedure. They were asked about their current relaxation state before HaptoRelax was applied to their left arm. They were instructed to place their arm on an elevated surface for comfort and to allow freedom of movement for the wrist springs. To eliminate bias and prevent visual observation of the device, a visual barrier was positioned in front of the wrist. Earplugs and headphones were provided to minimize distractions in the laboratory office setting. Participants were also instructed to close their eyes to enhance focus on the sensation. Once everything was prepared, a demonstration of the touch sensation was presented three consecutive times.

First, the two short SMA springs were actuated with Arduino for 7 seconds by 21.28W. Afterwards, the long SMA spring was actuated with a Power Supply for 10 seconds (7.5W). The trial was finished by a cooldown of the heated springs using a fan for 10 seconds.

4.4 Results

Screening Questionnaire

The screening questionnaire reveals information based on a sample of five participants, consisting of three males and two females, all within the age range of 22 to 25. Regarding their prior experience with virtual reality (VR), four participants reported having prior experience, while one participant indicated no prior experience with VR. In terms of familiarity with relaxation techniques, four participants reported having prior experience, whereas one participant stated no prior experience with such techniques.

At the beginning of the testing, participants rated their relaxation level on a scale of 1 to 5 (1 = "not at all relaxed, 5 = "extremely relaxed") with all choosing a 2 or 3, indicating a relatively low state of relaxation.

Part I: HaptoRelax without HMD

The average experience of using HaptoRelax without HMD was by 3.3 and ranged from 2 to 4 showing that some participants perceived it more positively than others but indicating an overall neutral opinion. Positive statements about the overall experience were that it feels like a warm massage chair and the warmth gives a positive feeling. Negatively perceived was the surprising wind coming from the fan for cooling the SMA springs down and the uncertainty of not knowing what will happen next as one participant stated.

The realism of the sensation was averagely rated with a 2.9 on 5 point Likert scale ranging from "very unreal" to "very real" showing a rather unreal touch experience. In terms of how the sensation felt it was said by three participants that the warmth felt comfortable. One participants though, rated the squeezing as artificial with a too-weak strength. Another participant draw similarities with a weak child grabbing the wrist. One participant didn't feel the contraction.

Post-device usage, participants rated their relaxation state at 3.4, indicating a slight increase. The gentle sensation was positively perceived as consoling by one participant. Another participant found relaxation by closing their eyes and focusing on the sensation. Warmth served as a relaxation factor for another participant. All participants mentioned a negative impact on relaxation due to the testing setup, including the office environment and the cooling fan.

Part II: HaptoRelax with HMD

Using HaptoRelax in VR was averagely rated with 4, ranging from 3.5 to 5. Feedback from two participants highlighted a gripping sensation with visual and auditory cues. However, one participant observed wrist compression in the VR scene, but only felt it on the side. Another participant noted a disparity between their actual posture and the VR representation.

Participants reported an average relaxation state of 3.6, ranging from 2 to 4.5, indicating a high level of relaxation post-usage. Positive feedback highlighted the device's ability to enhance the realism of the experience, along with the comfortable VR environment and immersive sound effects, such as bird chirping. Another participant appreciated the integration of multiple senses (touch, sight, and hearing). However, some participants mentioned imperfect synchronization with the scene and limited freedom of movement, with only one-dimensional deformation and less detailed device movement compared to a real human hand. They found it enjoyable, particularly in combination with augmented reality (AR).

Negative factors that disrupted relaxation included the presence of a fan and disorientation within the VR scene. Additionally, one participant found the virtual avatar unpleasant. Feedback suggested a desire for a stronger grabbing sensation and the absence of visual cues for the spring mechanism to enhance relaxation. While warmth was pleasant, it felt more meaningful with accompanying visual information. Overall, the experience was enjoyable, especially in

VR, but still fell short of replicating the realism desired. Stronger grabbing and improved mapping of motion to squeezing were suggested for enhancement.

Post Questionnaire

After the testing the participants were asked to fill out a short questionnaire evaluating the short VR sequence in terms of VR induced symptoms and effects. On a 7-point Likert scale from “absent” to 7 “extreme”, all participants stated an absent of very mild nausea, dizziness, fatigue, and discomfort. One participant stated an mild and one participant even an intense disorientation within the VR scene. The postural instability was very intense for the two participants.

5 USE CASE SCENARIO

5.1 VR Application for Intimate Connections

HaptoRelax can be used in a virtual environment where an intimate touch happens more likely, such as virtual spaces where people can meet their families or friends who are living remotely. VR applications such as [7, 17] let users explore tranquilizing VR scenes for the purpose of relaxation. By meeting people’s loved ones in these settings and exploring the scene together with HaptoRelax, they can not only visually touch each other, but also feel the warm contact, which can eventually elicit people to alleviate their stress in everyday remote society.

5.2 AR application for Meditation, Reflection, and Mindfulness

HaptoRelax can generate repetitive and calm grabbing on the wrist. By merging it with an AR application, HaptoRelax can be used as a meditation, reflection, and mindfulness guidance in everyday life. According to [1, 3], breathing guidance can improve one’s mental and physical health. As the slow pace of haptic action has a positive correlation with perceived calmness and relaxed breathing rate [19], the warm and gentle repetitive grabbing of HaptoRelax can be used with corresponding visual guidance in AR display. Its implementation into AR allows easier access to relaxation in everyday settings due to the see-through display of AR.

6 CONCLUSION

In conclusion, the results of our study shed light on the authenticity and effectiveness of HaptoRelax for simulating human touch and promoting relaxation.

To answer the first question, it can be stated that participants had varying experiences, with some positive and others mixed. The sensation’s authenticity received low ratings. With the HMD, participants rated the experience higher but noted discrepancies between visual and physical sensations. Post-usage, participants reported a slight increase in relaxation. Improvements are necessary to enhance the grabbing sensation’s realism and align physical and visual experiences in VR.

To answer the second question of our User Study on how effective HaptoRelax is to promote relaxation, it can be concluded that our device shows potential to promote relaxation. Participants reported a slight increase in relaxation post-usage. Positive feedback highlighted the consoling gentle sensation, relaxation through closing eyes and focusing, and the device’s warmth. However, negative factors such as the testing setup (cooling fan presence, VR scene disorientation) and limitations in functionality (weaker grabbing sensation) disrupted relaxation.

Overall, while HaptoRelax shows potential, further development is required to enhance touch authenticity and address negative factors impacting relaxation.

To enhance touch authenticity and relaxation effectiveness, improvements should focus on strengthening grabbing sensations to match a human grip’s strength. Enhancing synchronization between

visual representation and physical sensations in the VR scene is crucial to minimize disparities and delays, ensuring a seamless and realistic user experience.

In summary, HaptoRelax has potential, but advancements are needed to maximize authenticity, user experience, and overall effectiveness in providing an immersive and relaxing touch simulation.

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